

Birla Institute of Technology and Science, Pilani
First Semester, 2015-16
CHE F212, Fluid Mechanics

Quiz 3(Open Book)

26/08/15

15 min, 5 points

NAME:

ID:

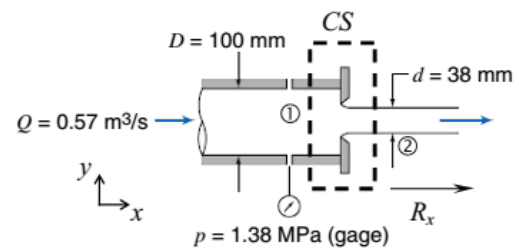
Section:

A flat plate orifice of 50 mm diameter is located at the end of a 100 mm diameter pipe. Water flows through the pipe and orifice at 0.57 m³/s. The diameter of the water jet downstream from the orifice is 38 mm.

a) Mention the assumptions you are considering for solving the problem [1.5]

b) Calculate the external force required to hold the orifice in place.

Neglect friction on the pipe wall. [3.5]



Given: Water flow through orifice plate.

Find: Force to hold plate.

Solution:

Basic equation: Momentum flux in x direction for the elbow $F_x = F_{S_x} + F_{B_x} = \frac{\partial}{\partial t} \int_{CV} u \rho dV + \int_{CS} u \rho \vec{V} \cdot d\vec{A}$

Assumptions: 1) Steady flow 2) Incompressible flow 3) Uniform flow

$$\text{Hence } R_x + p_{1g} \cdot A_1 - p_{2g} \cdot A_2 = V_1 \cdot (-\rho \cdot V_1 \cdot A_1) + V_2 \cdot (\rho \cdot V_2 \cdot A_2) \quad R_x = -p_{1g} \cdot A_1 + \rho \cdot (V_2^2 \cdot A_2 - V_1^2 \cdot A_1)$$

$$\text{From continuity} \quad Q = V_1 \cdot A_1 = V_2 \cdot A_2$$

$$\text{so } V_1 = \frac{Q}{A_1} = 0.57 \frac{\text{m}^3}{\text{s}} \times \frac{4}{\pi (0.1 \text{ m})^2} = 72.6 \frac{\text{m}}{\text{s}} \quad \text{and} \quad V_2 = V_1 \cdot \frac{A_1}{A_2} = V_1 \cdot \left(\frac{D}{d}\right)^2 = 72.6 \frac{\text{m}}{\text{s}} \times \left(\frac{100}{38}\right)^2 = 502.8 \frac{\text{m}}{\text{s}}$$

NOTE: Problem has an error: Flow rate should be 0.057 m³/s not 0.57 m³/s! We will provide answers to both.
Hence

$$R_x = -1.38 \times 10^6 \frac{\text{N}}{\text{m}^2} \times \frac{\pi (0.1 \text{ m})^2}{4} \times 999 \frac{\text{kg}}{\text{m}^3} \times \left[\left(502.8 \frac{\text{m}}{\text{s}} \right)^2 \times \frac{\pi (33 \text{ mm})^2}{4} - \left(72.6 \frac{\text{m}}{\text{s}} \right)^2 \times \frac{\pi (100 \text{ mm})^2}{4} \right] \times \left(\frac{1 \text{ m}}{1000 \text{ mm}} \right)^2 \times \frac{\text{N} \cdot \text{s}^2}{\text{kg} \cdot \text{m}}$$

$$R_x = 234282 \text{ N}$$

With more realistic velocities

Hence

$$R_x = -1.38 \times 10^6 \frac{\text{N}}{\text{m}^2} \times \frac{\pi (0.1 \text{ m})^2}{4} \times 999 \frac{\text{kg}}{\text{m}^3} \times \left[\left(50.3 \frac{\text{m}}{\text{s}} \right)^2 \times \frac{\pi (38 \text{ mm})^2}{4} - \left(72.6 \frac{\text{m}}{\text{s}} \right)^2 \times \frac{\pi (0.1 \text{ m})^2}{4} \right] \times \left(\frac{1 \text{ m}}{1000 \text{ mm}} \right)^2 \times \frac{\text{N} \cdot \text{s}^2}{\text{kg} \cdot \text{m}}$$

$$R_x = 7972 \text{ N}$$